

Searching Algorithms

Linear Search & Binary Search

Concept • Working • Examples • Time Complexity

Concept: Check Every Element

Array ke har element ko ek ek karke check karte hain — shuru se aakhir tak — jab tak target mil na jaye ya array khatam na ho jaye.

- 1 Index 0 se shuru karo
- 2 Har element ko target se compare karo
- 3 Match mile → index return karo
- 4 Array khatam ho jaye → "not found"

Works On

- ✓ Sorted arrays
- ✓ Unsorted arrays
- ✓ Any list / sequence

Time Complexity

$O(n)$

worst case

Example: Find the value 23



Index 0 → 5 (no match) → Index 1 → 8 (no match) → **Index 2 → 23 ✓ Found!**

3 comparisons taken to find the target at index 2.

Best case: target is the first element → $O(1)$. Worst case: target is last, or absent → $O(n)$.

Concept: Divide the Search Space

Yeh technique sirf sorted array par kaam karti hai. Har step par array ka aadha hissa discard kar diya jata hai.

- 1 Middle element ko check karo
- 2 Target chota hai → left half mein jao
- 3 Target bara hai → right half mein jao
- 4 Match milne tak, ya range khatam hone tak repeat

Precondition

Array MUST be sorted

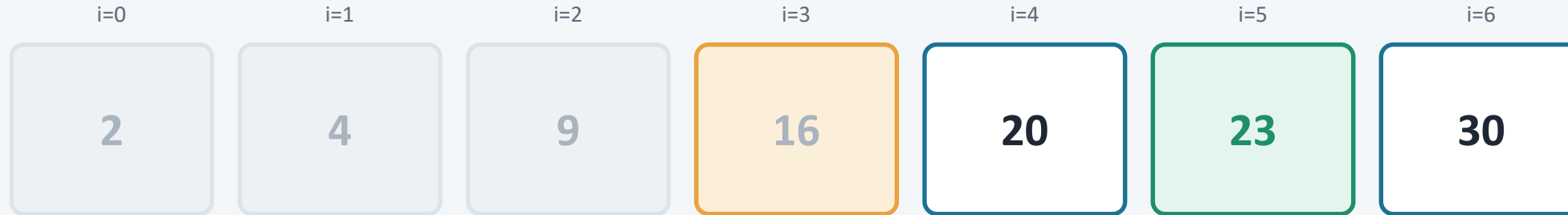
Warna middle-element comparison ka logic hi ghalat ho jata hai.

Time Complexity

$O(\log n)$

worst case

Example: Find 23 in a sorted array



Step 1:

low=0, high=6, mid=3 → value 16. $23 > 16$ → search right half

Step 2:

low=4, high=6, mid=5 → value 23 ✓ Found!

Sirf 2 steps mein target mil gaya — 7 elements ke liye linear search ko 6 steps lagtay.

COMPARISON

Time Complexity: Side by Side

	Linear Search	Binary Search
Best case	$O(1)$	$O(1)$
Worst case	$O(n)$	$O(\log n)$
Requires sorted array?	No	Yes
Extra space	$O(1)$	$O(1)$

For $n = 1,000,000$

~1,000,000

steps — Linear Search (worst case)

~20

steps — Binary Search (worst case)

Bara n ke liye binary search bohat fast hai — har step par search space aadha ho jata hai.

Key Takeaways

Linear Search

Simple, kisi bhi array (sorted ya unsorted) par chalta hai — lekin slow, $O(n)$.

$O(n)$

Binary Search

Fast, $O(\log n)$ — lekin array sorted hona zaroori hai.

$O(\log n)$

Rule of thumb: agar data sorted aur bara hai → Binary Search. Agar chota ya unsorted hai → Linear Search kaafi hai.